

TrES-5b

R-0745

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_241112 · created 2026-07-28T02:15:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

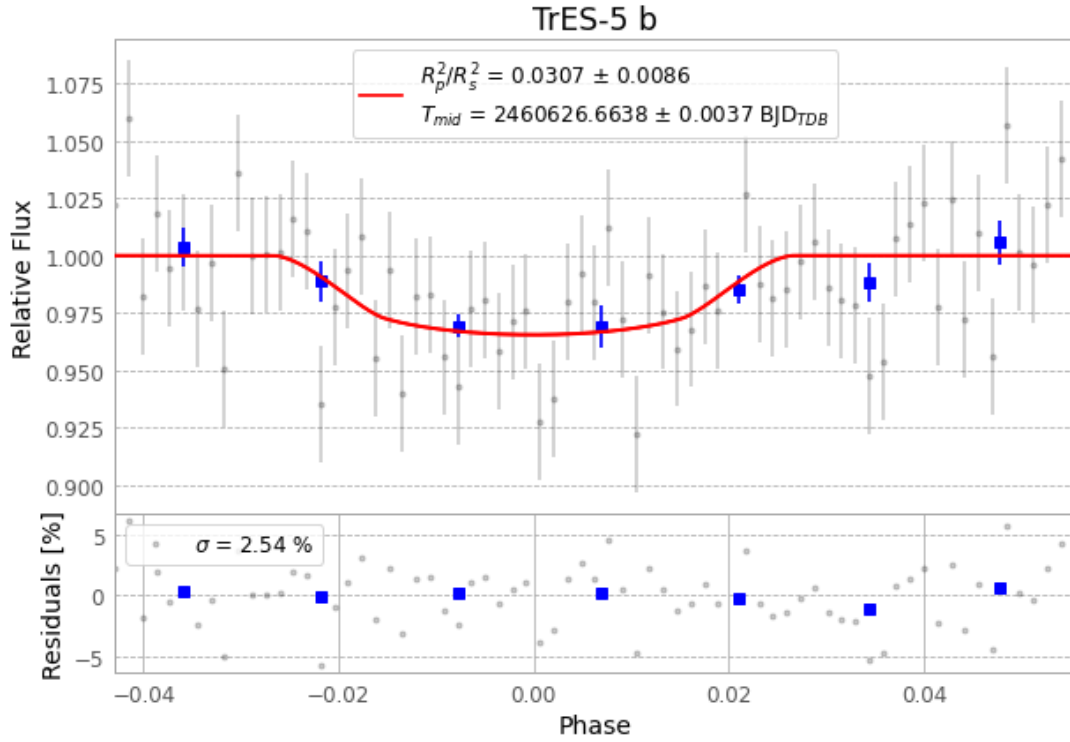
Results

Mid-Transit Time (Tmid)	2460626.6638 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.024
Transit depth (Rp/Rs)^2	3.07 +/- 0.86 [%]
Orbital Inclination (inc)	84.73 +/- 2.07
Airmass coefficient 1 (a1)	1.054 +/- 0.027
Airmass coefficient 2 (a2)	-0.034 +/- 0.016
Scatter in the residuals of the lightcurve fit is	2.54 %
Best Comparison Star	#1 - [348, 416]
Optimal Method	PSF photometry
Transit Duration (day)	0.077 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.175 ± 0.024 vs 0.142 (deviation 1.38σ)
Duration fit vs published	0.077 d vs 0.0758 d (deviation 0.1σ)
Mid-time O–C	-8.3 min
Depth SNR	7.3
Residual scatter	2.54 %

Light curve



Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 278], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4ea62a3240642130...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), TRES-5241112022115.FITS → TRES-5241112055115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-241112.log (212349f81a81...), FinalLightCurve_TrES-5 b_2024-11-11.png (24ca4b7b737c...), FinalLightCurve_TrES-5 b_2024-11-11.csv (7ccc0695bea0...)

Compute resources

Wall time 65.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:15Z.